

Staff Application Form for Experiments with Human Participants

Please check one:

- ☐ I need a formal approval letter for an external agency (TÜBİTAK, etc.)
- ☐ An internal communication letter informing me of the approval will be sufficient

1. Name of applicant (graduate students should indicate their supervisors):

İrem Damla Karagöz (Supervisor: Prof. Dr. Uğur Güdükbay)

2. Funder of grant/studentship if any:

N/A (Individual Research Project, CS490)

3. Full title of experiment/project:

Investigating Personality Perception through LLM-Generated 3D Gesture Profiles and Stick-Figure Animations

4. When do you wish to start data collection:

30 March 2026

5. Aims of project:

- This project researches how personality traits map to nonverbal gesture styles using a computational pipeline.
- The study aims to evaluate the ability of a Large Language Model (Google Gemini) to produce alternative gesture profiles based on eight normalized parameters (gesture_rate, gesture_amplitude, gesture_speed, gesture_symmetry, posture_openness, head_motion_rate, head_motion_amplitude, and idle_sway).
- The study compares baseline profiles against those generated by the LLM.
- By analyzing LLM consistency, proximity to baselines, and cross-condition separation, the project evaluates whether algorithmically generated nonverbal behaviors can reliably convey intended personality traits.

6. What will the participants have to do?

- Participants will participate in a perception study conducted via a computer interface.
- They will view a series of 6-second, 3D animations (GIFs) rendered at 24 fps.
- These animations visualize different personality conditions driven by 8-dimensional gesture vectors.

- For each animation, they will estimate the personality traits of the character using the Big Five (OCEAN) model, answer comparison questions, and provide a 1–5 confidence rating.
- The study is expected to take approximately **10–15 minutes**.

Safety Statement: The stimuli consist solely of neutral, computer-generated stick-figure renderings. They do not contain any offensive, graphic, violent, or emotionally disturbing material; therefore, participants will not be exposed to stimuli that may upset them.

7. What sort of people will the participants be and how will they be recruited?

- Participants will be adult volunteers (18+) recruited from the Bilkent University student community.
- Recruitment will be done through general announcements and peer communication.
- Bilkent University students who have the potential of being graded by the investigators will not participate in the study.
- Participation is voluntary, and no course credit or payment will be provided.

8. Arrangements for consent and debriefing:

- **Consent:** Before viewing stimuli, participants will be presented with an information sheet and an informed consent form on-screen. Participation will only begin after the participant explicitly indicates consent by pressing a designated key.
- **Debriefing:** At the end of the session, a debriefing screen will explain that their ratings help validate the efficacy of LLM-generated nonverbal behavior in artificial agents.

9. How will you guarantee the confidentiality of participants?

- No personally identifying information (names or student IDs) will be collected.
- Data will be organized using random alphanumeric codes.
- All data will be stored on password-protected devices and will only be accessible to the research team.
- Results will be reported in an aggregated statistical format (e.g., mean consistency scores and L2 distance averages), ensuring no individual can be identified.

Signature(s):

Person carrying out the work: _____ (**İrem Damla Karagöz**)

Supervisor: _____ (**Prof. Dr. Uğur Güdükbay**)

Appendix: Sample Questions for Tasks

- Which person appears more outgoing, sociable, and energetic?
- Which person appears more friendly, cooperative, and warm?

- Which person appears more organized, responsible, and reliable?
- Which person appears more calm, emotionally stable, and resilient?
- Which person appears more open to new experiences, creative, and curious?